

Sanity Checks for Elicitation of Self Preference in LLM Evaluators

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Abstract

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1. Introduction

As language models see increasing deployment as evaluators, raters, and proxies for human preference, researchers have observed consistent patterns of unreliability (Gu et al., 2025; Li et al., 2024). These patterns are frequently categorized as *biases* (Dietz et al., 2025; Ye et al., 2024) and are often interpreted through anthropomorphic lenses or linked to emerging theory-of-mind frameworks. To index these reliability gaps, current literature has identified a gamut of behaviors, some of which—namely, the auto-evaluation behaviors described as *self-preference* (Panickssery et al., 2024) or *narcissism* (Liu et al., 2024)—suggest that models exhibit *situational awareness* (Laine et al., 2023): that is, the ability to infer details about the evaluation context (e.g., recognizing their own output) and modify their behavior accordingly to favor their own generations.

Ironically, eliciting of awareness behaviors like self-preference remains as unpredictable as reliable evaluation itself. While self-preference is often presented as a failure of alignment, existing metrics frequently lack robust null hypotheses. Early testing omitted essential “ground-truth” baselines, leading researchers to claim self-preference even when a model’s high ratings were simply an accurate assessment of its own superior performance (Chen et al., 2025a; Wataoka et al., 2025). While recent efforts have introduced baselines for generation quality, a critical gap remains regarding *evaluation quality*: the ability of a judge model to assign accurate preference ratings regardless of its own generative capacity. In the context of scalable oversight, the inability to distinguish between a model’s general evaluative

proficiency and an active self-bias undermines the validity of situational awareness claims.

In this paper, we argue that claims of situational awareness in LLM evaluators require observed, statistically significant effects under more rigorous null hypotheses. Using the case study of self-preference, we propose baselines for two under-explored dimensions: first, a model’s baseline effectiveness in making superiority judgements on difficult problems (judgement skill); and second, the direct correlation between a model’s ability to recognize its own outputs and its tendency to favor that output (self-recognition).

Formalizing these tests, we find that many evaluation harnesses which claim to elicit self-preference fail to reject one or both null hypotheses. This suggests that previously reported self-preference effects may be confounded by general evaluation patterns—such as behavior under uncertainty and weak oversight dynamics—rather than an affirmative, deliberate composition of self-preference behavior. Furthermore, we show that recent discoveries of “self-preference” (Roytburg et al., 2025) or “self-recognition” (Ackerman & Panickssery, 2025) vectors in activation space erroneously persist in evaluation settings that do not have a “self” to prefer or recognize.

Our contributions are as follows:

1. We formalize two null hypotheses to rigorously test claims of self-preference in language model evaluators.
2. We empirically evaluate these hypotheses across multiple experimental setups from recent literature, revealing that many prior claims of self-preference fail to reject these null hypotheses.
3. We investigate the mechanistic underpinnings of self-preference through representation engineering, assessing the efficacy of steering vectors in modulating self-recognition and self-preference behaviors.

Together, these findings offer a more realistic view of prevailing theory-of-mind initiatives; with stronger statistical baselines we can produce cleaner signals of situational awareness, a precursor to reliable scalable oversight strategies.

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2. Background

Literature on LLM evaluation has rapidly expanded in recent years, with numerous surveys cataloging the diverse methodologies and challenges (Zheng et al., 2023; Shen et al., 2023; Chen et al., 2024; Li et al., 2024; 2025; Gu et al., 2025). A significant subset of this work focuses on the reliability and biases of LLM evaluators, identifying various failure modes including inconsistency, susceptibility to manipulation, and context sensitivity (Koo et al., 2024; Ye et al., 2024; Dietz et al., 2025).

Among these, *self-preference* has emerged as a particularly intriguing phenomenon, where models appear to favor their own outputs over those of other models. Panickssery, Bowman and Feng (2024) isolate this behavior using proxy tests on summarization tasks, while Liu, Moosavi and Lin (2024) observe a similar behavior using judges trained on encoding-based representations. This behavior has been interpreted as evidence of *situational awareness*, suggesting that models can recognize their own outputs and adjust their evaluations accordingly (Laine et al., 2023).

3. Evolving Baselines

3.1. Setup

First, consider a judge model J prompted to conduct pairwise evaluations task responses completed by itself and another generative model R on a given task x . We define outputs from a judge model J as o_J and outputs from the reference model R as o_R . The judge model is shown x and the two outputs and returns a number between $[0, 1]$ corresponding to the probability that J generates a token corresponding to a vote for itself: $S_J(x, o_J, o_R) : (\mathcal{X}, \mathcal{O}_J, \mathcal{O}_R) \rightarrow [0, 1]$. Then, taken over all tasks $x \in \mathcal{X}$, we can define the self-preference score (**SP**) of model J against model R as:

$$\mathbf{SP}(J, R, \mathcal{X}) = \mathbb{E}_{(x, o_J, o_R)}[S_J(x, o_J, o_R)]$$

where a score greater than 0.5 indicates a preference for its own output.

3.2. Comparison against “Generation Quality”

The metric above, introduced by Panickssery, Bowman and Feng (2024) to elicit self-preference behavior, harbors a key limitation: the judge model J may simply be better at the task than the reference model R , leading to a higher self-preference score that reflects evaluation skill rather than true self-bias. In response, different approaches have introduced a formulation which relies on oracle labels which resemble alignment with human preferences (Wataoka et al., 2025; Chen et al., 2025b) or verifiable outputs (Chen et al., 2025a).

Now, consider ground truth labels: $G(x, o_J, o_R) : (\mathcal{X}, \mathcal{O}_J, \mathcal{O}_R) \rightarrow \{0, 1\}$ produced by some oracle G which

takes the same pairwise input and returns the indicator $\mathbb{I}[o_J \succ o_R]$. The win-rate, or *generation quality*, of the judge is defined as the expected value of the oracle label:

$$\mathbf{Acc}(J, R, \mathcal{X}) = \mathbb{E}_{(x, o_J, o_R)}[G(x, o_J, o_R)]$$

It follows that the optimal self-preference rate is when $\mathbf{SP}(J, R, \mathcal{X})$ is perfectly aligned with $\mathbf{Acc}(J, R, \mathcal{X})$. Then, the self-preference bias of a judge J may be defined as this gap:

$$\mathbf{Bias}(J, R, \mathcal{X}) = \mathbf{SP}(J, R, \mathcal{X}) - \mathbf{Acc}(J, R, \mathcal{X})$$

3.3. Illegitimate versus Legitimate Self-Preference

The above formulation enables us to frame **Bias** as the mean self-preference probability derived in Section 3.1 subtracted by the expected winrate, thus successfully decoupling a model’s “generation quality” from its self preference.

oracle labels also enable us to discern whether self-preference is valid for any individual example. We then define the *illegitimate* self preference rate (**ILSP**) and the *legitimate* self-preference rate (**LSP**):

$$\mathbf{ILSP}(J, R, G, \mathcal{X}) = \mathbb{E}_{(x, o_J, o_R)}[S_J(x, o_J, o_R) | \neg G(x, o_J, o_R)]$$

$$\mathbf{LSP}(J, R, G, \mathcal{X}) = \mathbb{E}_{(x, o_J, o_R)}[S_J(x, o_J, o_R) | G(x, o_J, o_R)]$$

Illegitimate self-preference rates are taken from “losing” examples per oracle labels. By definition, the human-aligned ILSP would match the oracle labels at exactly 0. On the contrary, **legitimate** self-preference rates are taken from “winning examples”, and the the human-aligned LSP would be exactly 1.

We can reformulate bias as the weighted average of these terms:

$$\mathbf{Bias} = ((1 - \mathbf{Acc}) \cdot \mathbf{ILSP}) - (\mathbf{Acc} \cdot (1 - \mathbf{LSP}))$$

3.4. Hypothesis 1: “Evaluator Quality” Baseline

Under this framing, a new failure mode comes to light. Most neural networks that return a softmax in logit space will not be trained to produce exact 0 or 1 answers. Thus, for the **ILSP** case, any surplus in probability (uncertainty) is automatically penalized unless that uncertainty is counterbalanced by an equal deficit in probability in the **LSP** case.

Then [a little bit more theory... idk...], yada yada yada.

Thus, we propose the *Evaluator Quality Baseline*. Consider now a proxy model K , and take a subset of the dataset $\mathcal{X}_K$ such that K and J are of *similar capability* against reference model R on $\mathcal{X}_K$ according to the oracle G . That is,

$$\mathcal{X}_K : \{x \in \mathcal{X} | \mathbb{I}[G(x, o_J, o_R) = G(x, o_K, o_R)]\}$$

Then, run a test returning a proxy preference score, where the judge evaluates the proxy K against the reference R instead of itself:

$$PP = \mathbb{E}_{x \in \mathcal{X}_K} [S_J(x, o_K, o_R)]$$

–**SP**($J, R, G, \mathcal{X}_K$). Then, the distribution of vote labels from J comparing J against R significantly different from J comparing a similarly-capable model K against R ? (H₀: $J(J \text{ v. } R) = J(K \text{ v. } R)$)

3.5. Hypothesis 2: Self-Recognition Baseline

The Self-Recognition Correlation: On examples where J exhibits bias toward J relative to both the ground-truth G and the control K , are these instances correlated with the model’s ability to recognize itself? (H₀: $\text{Corr}(J_{\text{pref}}, J_{\text{rec}}) = 0$)

3.6. Hypothesis 3: Mechanistic Insolvency Baseline

2) We show that many previous experimental setups fail to reject one or both of these null hypotheses. In many cases, evaluation setups previously thought to elicit self-preference do not meaningfully distinguish between judges evaluating themselves and judges evaluating others. Furthermore, we find a lack of correlation with self-recognition, a necessary precursor for intentional situational awareness.

3) Mechanistically, we evaluate the utility of representation engineering in mitigating these behaviors. Building on recent findings that identify a “self-authorship” vector in the residual stream (Ackerman & Panickssery, 2024), we attempt to use similar steering vectors to suppress self-recognition and, by extension, self-preference. We demonstrate that while these vectors can successfully modulate explicit authorship claims on in-distribution data, they fail to generalize in out-of-distribution (OOD) settings. Specifically, when J evaluates a model K with similar architectural or stylistic priors, the steering vector fails to decouple “preference” from “recognition.”

3.7. Hypothesis Testing

4. Experimental Details

4.1. Selected Setups

4.1.1. “DEBIASED GOLD JUDGES”

4.1.2. CROWDSOURCED ANNOTATIONS

4.1.3. VERIFIABLE OUTPUTS

4.2. Implementation

5. Results and Discussion

6. Limitations

7. Conclusion and Future Work

Impact Statement

Authors are **required** to include a statement of the potential broader impact of their work, including its ethical aspects and future societal consequences. This statement should be in an unnumbered section at the end of the paper (co-located with Acknowledgements – the two may appear in either order, but both must be before References), and does not count toward the paper page limit. In many cases, where the ethical impacts and expected societal implications are those that are well established when advancing the field of Machine Learning, substantial discussion is not required, and a simple statement such as the following will suffice:

“This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.”

The above statement can be used verbatim in such cases, but we encourage authors to think about whether there is content which does warrant further discussion, as this statement will be apparent if the paper is later flagged for ethics review.

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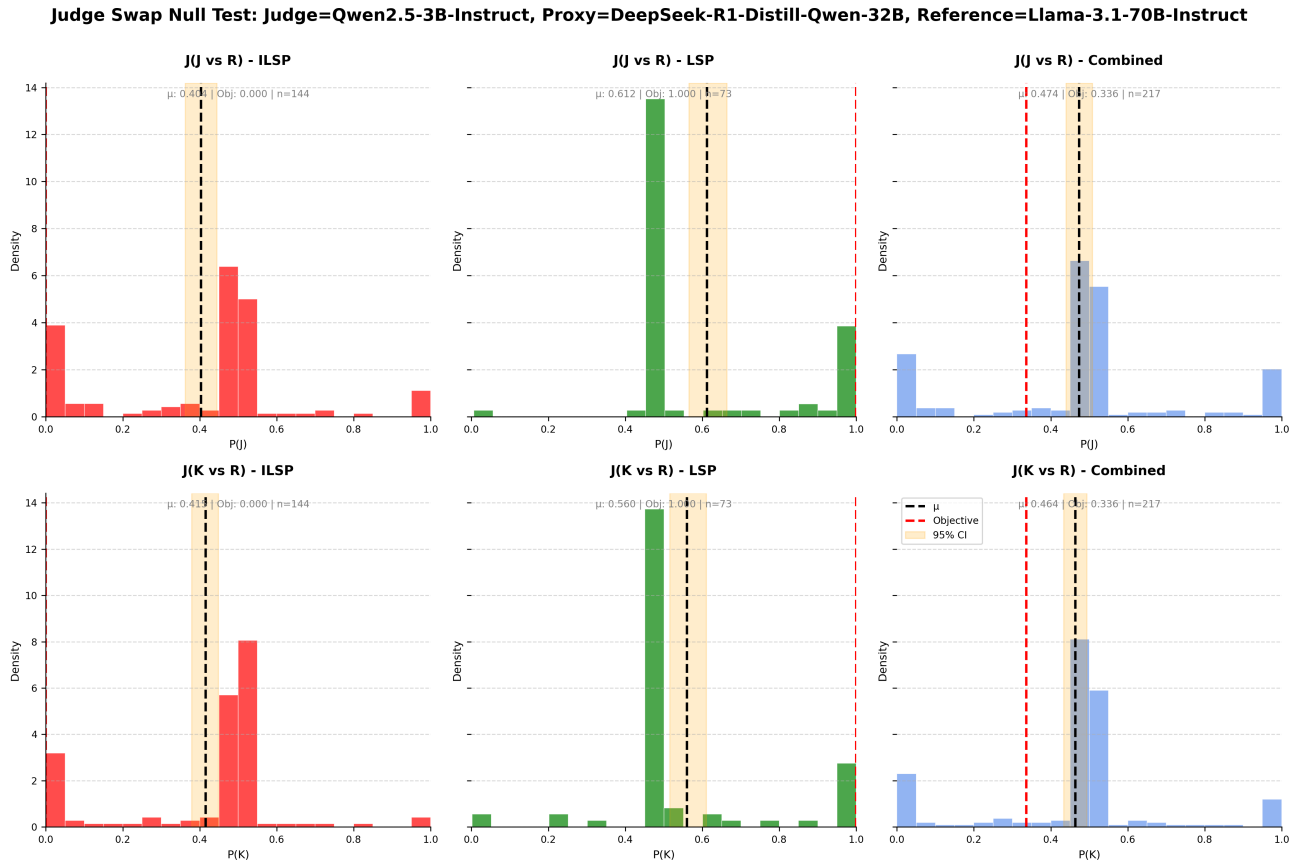


Figure 1. hehehe

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